b

miércoles, 30 de setiembre de 2026 19:14

Cinemática Inversa

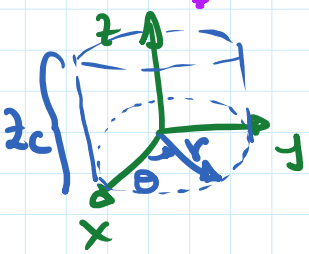
Métodos Geométricos:

Localización { orient. / posición } → Función No Lineal. → $\begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_n \end{bmatrix}$

Scara 3gdl [G1 - 1 1 1 S1] [RRP]

$x = L_1 \cos\theta_1 + L_2 \cos(\theta_1 + \theta_2)$
$y = L_1 \operatorname{sen}\theta_1 + L_2 \operatorname{sen}(\theta_1 + \theta_2)$
$z = -d_3.$

Int. → SCARA.



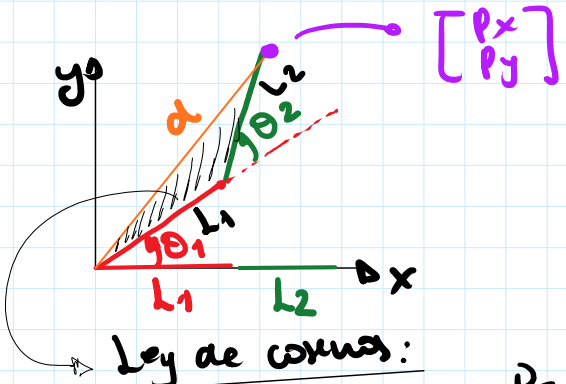
Relación Coord. Cilíndricas: $z_c = z$

Coord. Cartesianas

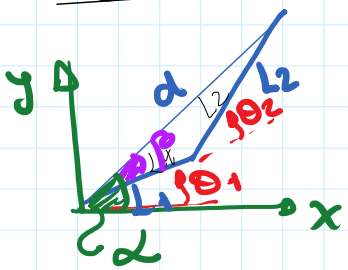
$\begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix}$ → Cinemática Inversa Scara 3gdl G1-1111S → $\begin{bmatrix} \theta_1 \\ \theta_2 \\ d_3 \end{bmatrix}$

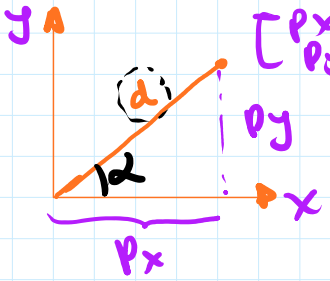
Solución

$d_3 = p_z.$



Ley de cosenos:





$d = \sqrt{p_x^2 + p_y^2}$

$\alpha = \tan^{-1}\left(\frac{p_y}{p_x}\right)$

Para θ_2:

$d^2 = L_1^2 + L_2^2 + 2L_1L_2C_2$

$C_2 = \frac{d^2 - L_1^2 - L_2^2}{2L_1L_2}$; $S_2^2 + C_2^2 = 1$

$S_2 = \pm\sqrt{1 - C_2^2}$

$\theta_2 = \tan^{-1}\left(\frac{S_2}{C_2}\right)$ →
- $\theta_2 = \tan^{-1}\left(+\frac{S_2}{C_2}\right)$
- $\theta_2 = \tan^{-1}\left(-\frac{S_2}{C_2}\right)$
- $\theta_2 = -\tan^{-1}(S_2/C_2)$

θ_1:

(1) ... $\theta_1 = \alpha - \beta$; $\alpha = \tan^{-1}\left(\frac{p_y}{p_x}\right)$

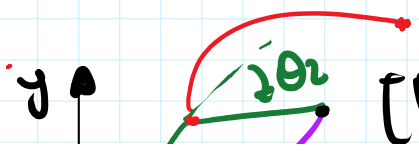
$L_2^2 = d^2 + L_1^2 - 2L_1 d C\beta$

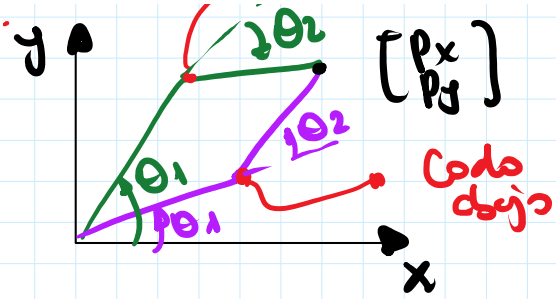
$C\beta = \frac{d^2 + L_1^2 - L_2^2}{2L_1 d}$; $S\beta = \pm\sqrt{1 - C\beta^2}$

β → $\beta = \tan^{-1}\left(+\frac{S\beta}{C\beta}\right)$; $\beta = -\tan^{-1}(S\beta/C\beta)$

En (1):
$\theta_1 = \alpha - \beta$
$\theta_1 = \alpha - (-\beta) = \alpha + \beta$

Codo arriba.





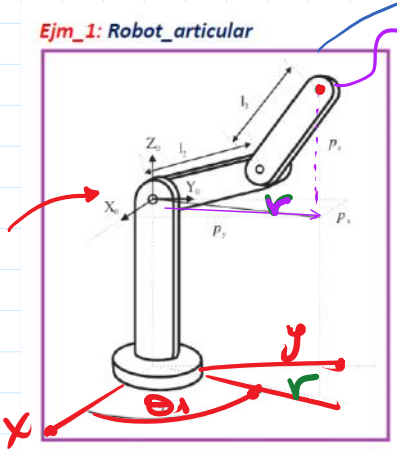
Resumen:

$$\theta_1 = \begin{cases} \alpha-\beta \\ \alpha-\beta \end{cases} \qquad \theta_2 = \begin{cases} \theta_2 \\ -\theta_2 \end{cases}$$

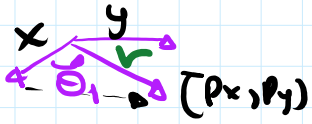
L_1, L_2, Seno, Coseno.

$$d_3 = p_z$$

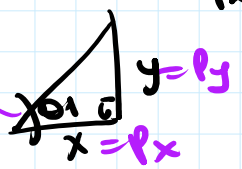
3gdl (Articulado)

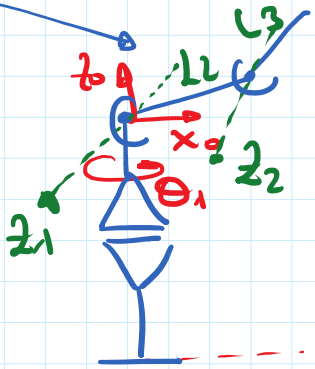


$\begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix}$



$$\theta_1 = \tan^{-1}\left(\frac{p_y}{p_x}\right)$$





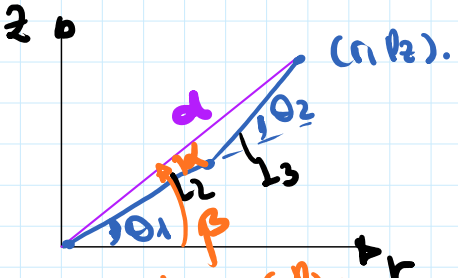
$$r = \sqrt{p_x^2 + p_y^2}$$

$$d = \sqrt{r^2 + p_z^2}$$

$$d^2 = L_2^2 + L_3^2 + 2L_2L_3C_2$$

$$C_2 = \frac{d^2 - L_2^2 - L_3^2}{2L_2L_3} \quad ; \quad S_2 = \pm\sqrt{1-C_2^2}$$

$$\theta_2 = \tan^{-1}\left(\frac{\pm S_2}{C_2}\right)$$



$$\beta = \tan^{-1}\left(\frac{p_z}{r}\right)$$

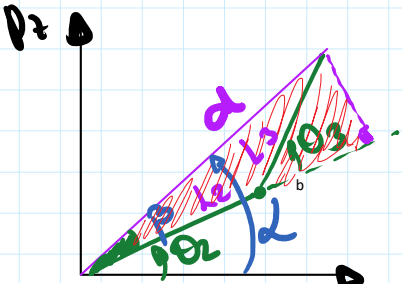
$$L_3^2 = L_2^2 + d^2 - 2L_3 d\, C\beta$$

$$C\beta = \frac{L_2^2 + d^2 - L_3^2}{2L_3 d}$$

$$S\beta = \pm\sqrt{1-C\beta^2}$$

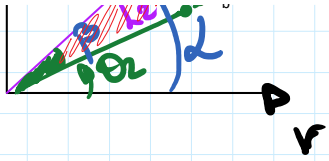
$$\beta = (\pm)\tan^{-1}\left(\frac{S\beta}{C\beta}\right)$$

$$\theta_1 = \alpha - \beta$$



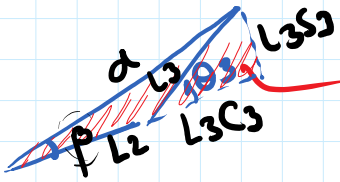
$$\theta_2 = \alpha - \beta \qquad \alpha = \tan^{-1}\left(\frac{p_z}{r}\right)$$

L_3S_3



$l_2 = 10$

$l_3 = 10$



$\tan\beta = \frac{L_3S_3}{L_2 + L_3C_3}$

$\therefore \theta_2 = \tan^{-1}\left(\frac{p_z}{\sqrt{p_x^2+p_y^2}}\right) - \frac{L_3S_3}{L_2+L_3C_3}$

Finalmente

$$q_2 = arctg\left(\frac{p_z}{\pm\sqrt{p_x^2 + p_y^2}}\right) - arctg\left(\frac{l_3 \operatorname{sen} q_3}{l_2 + l_3 \cos q_3}\right)$$